

## Fall 25, 4100 Problem Set 6, Hayden Fuller

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Given  $A = \begin{bmatrix} 2 & -4 & 1 & 3 \\ 3 & -6 & 0 & -3 \\ -1 & 2 & 1 & 0 \\ 4 & -8 & -1 & 3 \end{bmatrix}$  compute  $\text{adj}(A)$  and use it to find bases for  $\text{null}(A)$  and  $\text{null}(A^T)$

$$\text{adj}(A) = \begin{bmatrix} -36 & 0 & 72 & 36 \\ -18 & 0 & 36 & 18 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\det(A) = 0$ ,  $\text{rank}(A) = 3$ ,  $\dim \ker(A) = 1$ ,  $\dim \ker(A^T) = 1$   
basis vectors:

$\ker(A) = \text{span}\{(2, 1, 0, 0)^T\}$ ,  $\ker(A^T) = \text{span}\{(-1, 0, 2, 1)^T\}$   
 $A(2, 1, 0, 0)^T = 0$ ,  $A^T(-1, 0, 2, 1)^T = 0$

$$\text{adj}(A) = 18 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} -1 & 0 & 2 & 1 \end{bmatrix}$$

2. Find the eigenvalues and eigenvectors given the following matrices:

a.  $A = \begin{bmatrix} 4 & 3 \\ 5 & 6 \end{bmatrix}$

$$\det(A - \lambda I) = (4 - \lambda)(6 - \lambda) - 15 = \lambda^2 - 10\lambda + 9$$

$$\lambda = 9, \lambda = 1$$

$$(A - 9I)v = 0, v = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$(A - 1I)v = 0, v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

b.  $B = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 4 \\ 0 & 2 & -2 \end{bmatrix}$

$$\det(B - \lambda I) = (1 - \lambda)(\lambda^2 + 2\lambda - 8) = (1 - \lambda)(\lambda - 2)(\lambda + 4)$$

$$\lambda = 1, \lambda = 2, \lambda = -4$$

$$(B - 1I)v = 0, v = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$(B - 2I)v = 0, v = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

$$(B + 4I)v = 0, v = \begin{bmatrix} 3 \\ -5 \\ 5 \end{bmatrix}$$

3. Let  $T \in \mathcal{L}(\mathbb{C}^n, \mathbb{C}^n)$  be such that for linear independent  $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{C}^n$  we have that  $T\mathbf{v}_1 = \mathbf{v}_1 + 6\mathbf{v}_2$  and  $T\mathbf{v}_2 = 3\mathbf{v}_1 + 4\mathbf{v}_2$ . Find two eigenvalues and associated eigenvectors of  $T$ .

$$M = \begin{bmatrix} 1 & 3 \\ 6 & 4 \end{bmatrix}$$

$$\det(M - \lambda I) = (1 - \lambda)(4 - \lambda) - 18 = \lambda^2 - 5\lambda - 14 = 0$$

$$\lambda = 7, \lambda = -2$$

$$(M - 7I)v = 0, v = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$1v_1 + 2v_2, 7$$

$$(M + 2I)v = 0, v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$1v_1 - 1v_2, -2$$

$$T(v_1 + 2v_2) = 7(v_1 + 2v_2)$$

$$T(v_1 - v_2) = -2(v_1 - v_2)$$

4. Let  $A$  and  $B$  be similar such that  $A = M^{-1}BM$ . Prove that  $A$  and  $B$  have the same characteristic and minimal polynomial.

let  $A = M^{-1}BM$

then  $\chi_A(\lambda) = \det(\lambda I - A) = \det(\lambda I - M^{-1}BM) = \det(M^{-1}(\lambda I - B)M)$

$\det(M^{-1}(\lambda I - B)M) = \det(M^{-1})\det(\lambda I - B)\det(M) = \det(\lambda I - B) = \chi_B(\lambda)$

thus  $\chi_A(\lambda) = \chi_B(\lambda)$

let  $m_B(\lambda)$  be the minimal polynomial of  $B$

consider  $m_B(A)$

$m_B(A) = m_B(M^{-1}BM) = M^{-1}m_B(B)M$

but  $m_B(B) = 0$ , so  $m_B(A) = M^{-1}0M = 0$

hence  $m_B$  is also an annihilating polynomial for  $A$ , so the minimal polynomial  $m_A$  of  $A$  divides  $m_B$

by symmetry,  $m_B$  of  $B$  divides  $m_A$

two monic polynomials that divide each other must be equal, so  $m_A = m_B$

5. Find the set of all  $A \in \mathbb{R}^{2 \times 2}$  such that  $A^6 = I$  up to similarity classes by finding all possible minimal polynomials.

$m_A(x)$  divides  $x^6 - 1$

$$x^6 - 1 = (x^3 - 1)(x^3 + 1) = (x - 1)(x^2 + x + 1)(x + 1)(x^2 - x + 1)$$

possible minimal polynomials  $m(x) = x - 1, x + 1, x^2 + x + 1, x^2 - x + 1, x^2 - 1$

$m(x) = x - 1$ , then  $A = I$

$m(x) = x + 1$ , then  $A = -I$

$m(x) = x^2 - 1 = (x - 1)(x + 1)$ , then  $A$  is diagonalizable with eigenvalues 1 and -1

$$A \sim \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$m(x) = x^2 + x + 1$ , then

$$A \sim \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}, \text{ satisfying } A^3 = I \text{ and } A \neq I, \text{ so } A^6 = I$$

$$m(x) = x^2 - x + 1, \text{ then } A \sim \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}, \text{ satisfying } A^6 = I \text{ and } A^3 \neq I$$

$$A \in \{I, -I, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}, \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}\}$$

6. Let  $A \in \mathbb{R}^{n \times n}$  be invertible and have characteristic polynomial of  $p(\lambda)$ . Show that  $A^{-1}$  will have characteristic polynomial of

$$q(\lambda) = \frac{(-\lambda)^n}{|A|} p\left(\frac{1}{\lambda}\right)$$

$$p\left(\frac{1}{\lambda}\right) = \det\left(\frac{1}{\lambda}I - A\right) = \det\left(\frac{1}{\lambda}(I - \lambda A)\right) = \left(\frac{1}{\lambda}\right)^n \det(I - \lambda A)$$

$$\text{since } \det(I - \lambda A) = (-1)^n \det(\lambda A - I),$$

$$\text{this gives } p\left(\frac{1}{\lambda}\right) = \frac{(-1)^n}{\lambda^n} \det(\lambda A - I)$$

$$\text{hence } \det(\lambda A - I) = (-\lambda)^n p\left(\frac{1}{\lambda}\right)$$

$$\det(\lambda I - A^{-1}) = \det(A^{-1}) \det(\lambda A - I) = \frac{1}{\det(A)} (-\lambda)^n p\left(\frac{1}{\lambda}\right)$$

$$\text{therefore } \det(\lambda I - A^{-1}) = \frac{(-\lambda)^n}{\det(A)} p\left(\frac{1}{\lambda}\right)$$

QED